

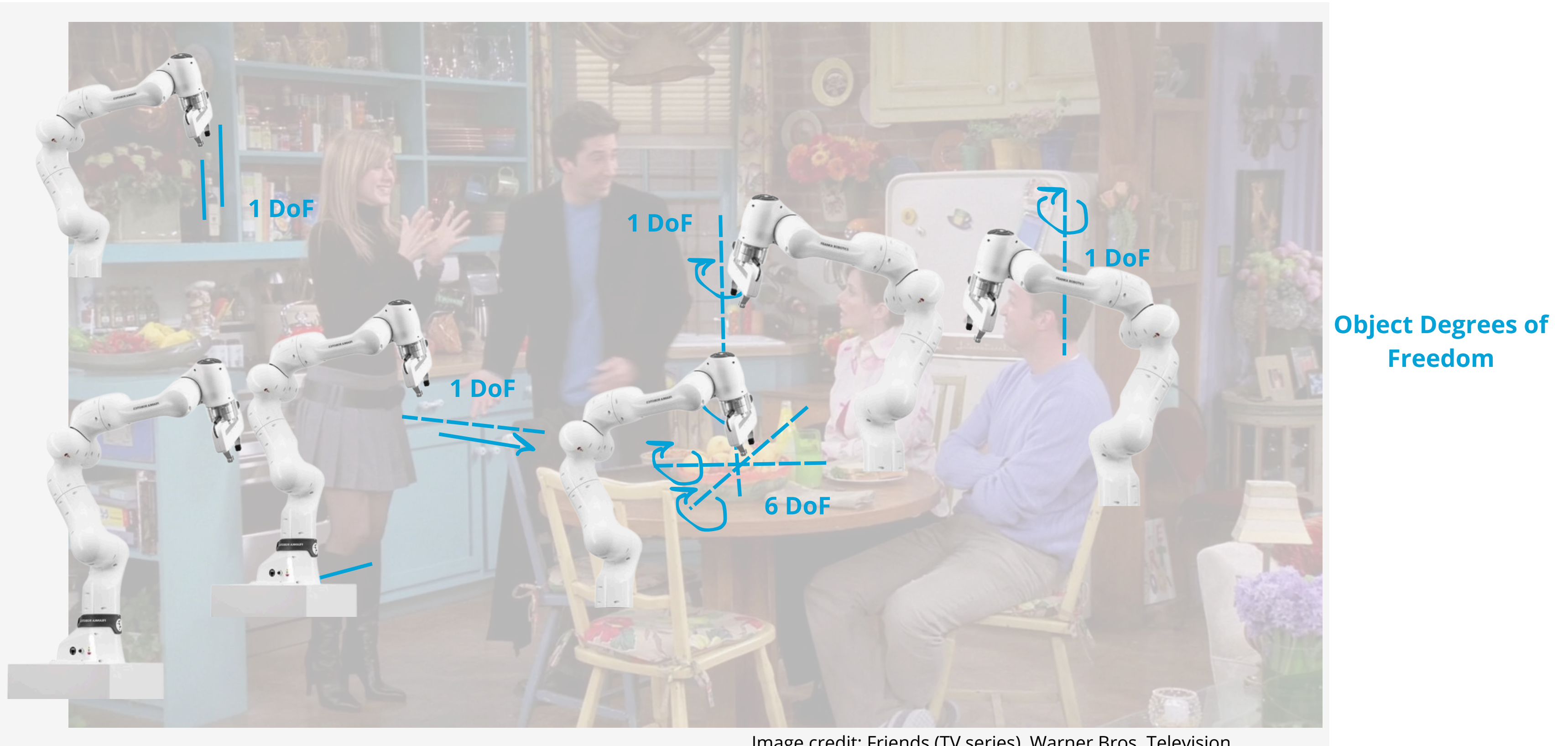
QDTraj: Exploration of Diverse Trajectory Primitives for Articulated Objects Robotic Manipulation

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How can a robot learn to manipulate articulated objects ?



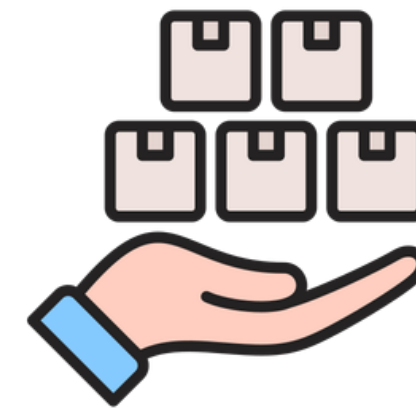
Challenge of learning robotic manipulation

Stable grasp + motion = **PREHENSILE manipulation**



Reinforcement Learning

Grasping → NO DENSE REWARD
no reward = no learning



Imitation Learning

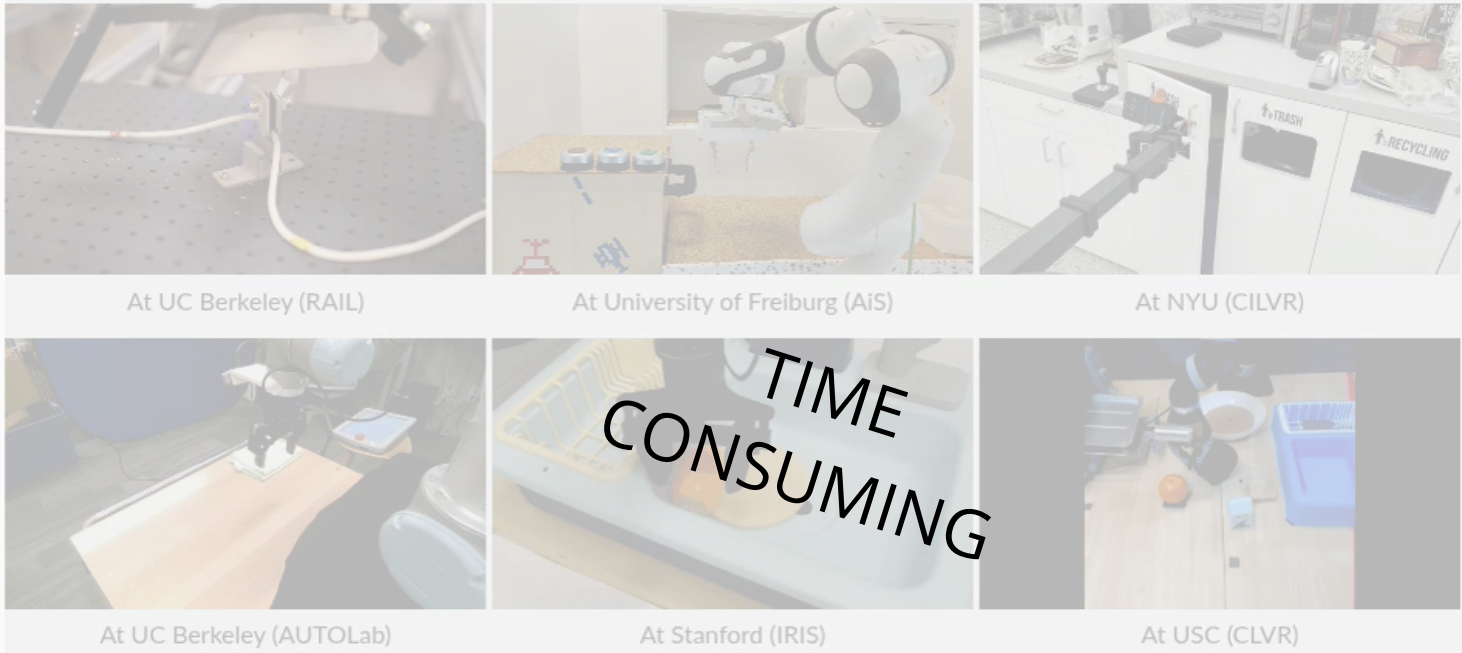
Requires a large number of demonstrations

Difficulty of expert data acquisition

Open-ended environment

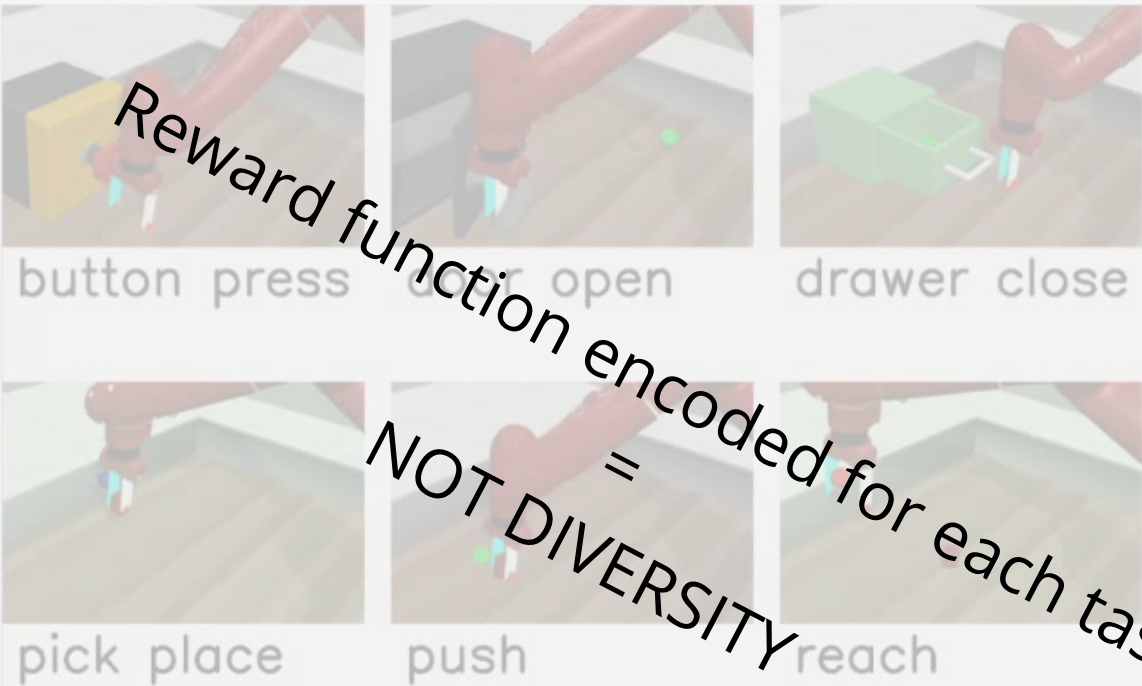


Teleoperated expert data



[1] Open X-Embodiment Collaboration. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. arXiv:2310.08864, 2023.

Simulated expert data



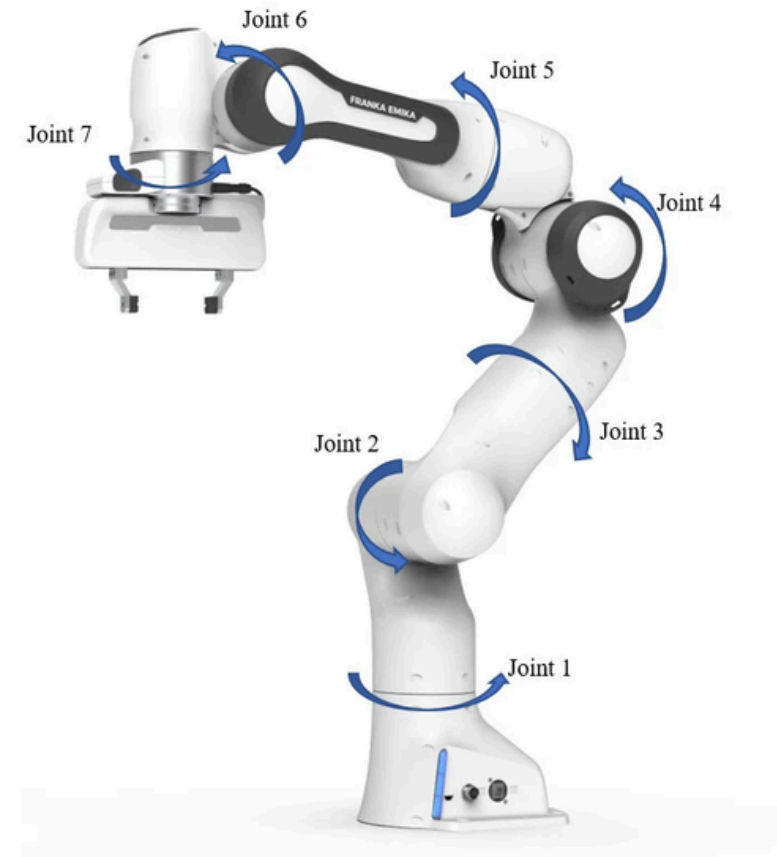
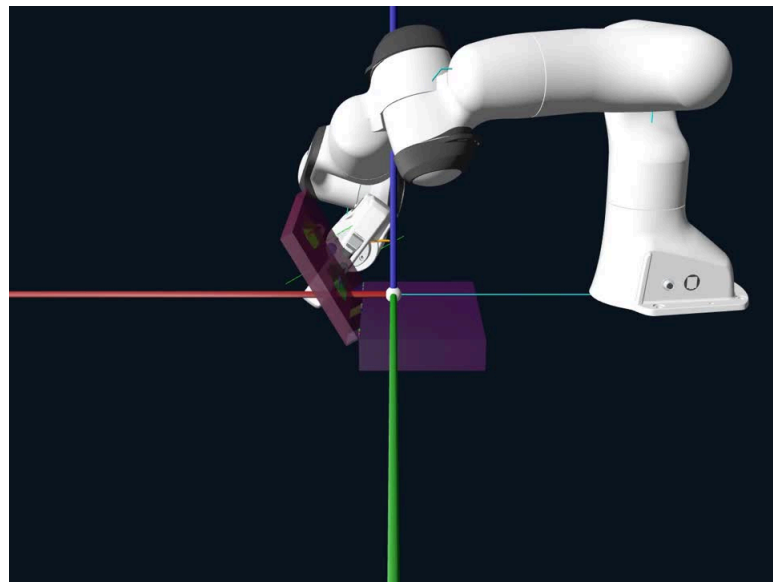
[2] Yu et al. "Meta-World: A Benchmark and Evaluation for Multi-Task and Meta Reinforcement Learning", arXiv:1910.10897, 2019.

IMPORTANCE OF NOVELTY AND DIVERSITY

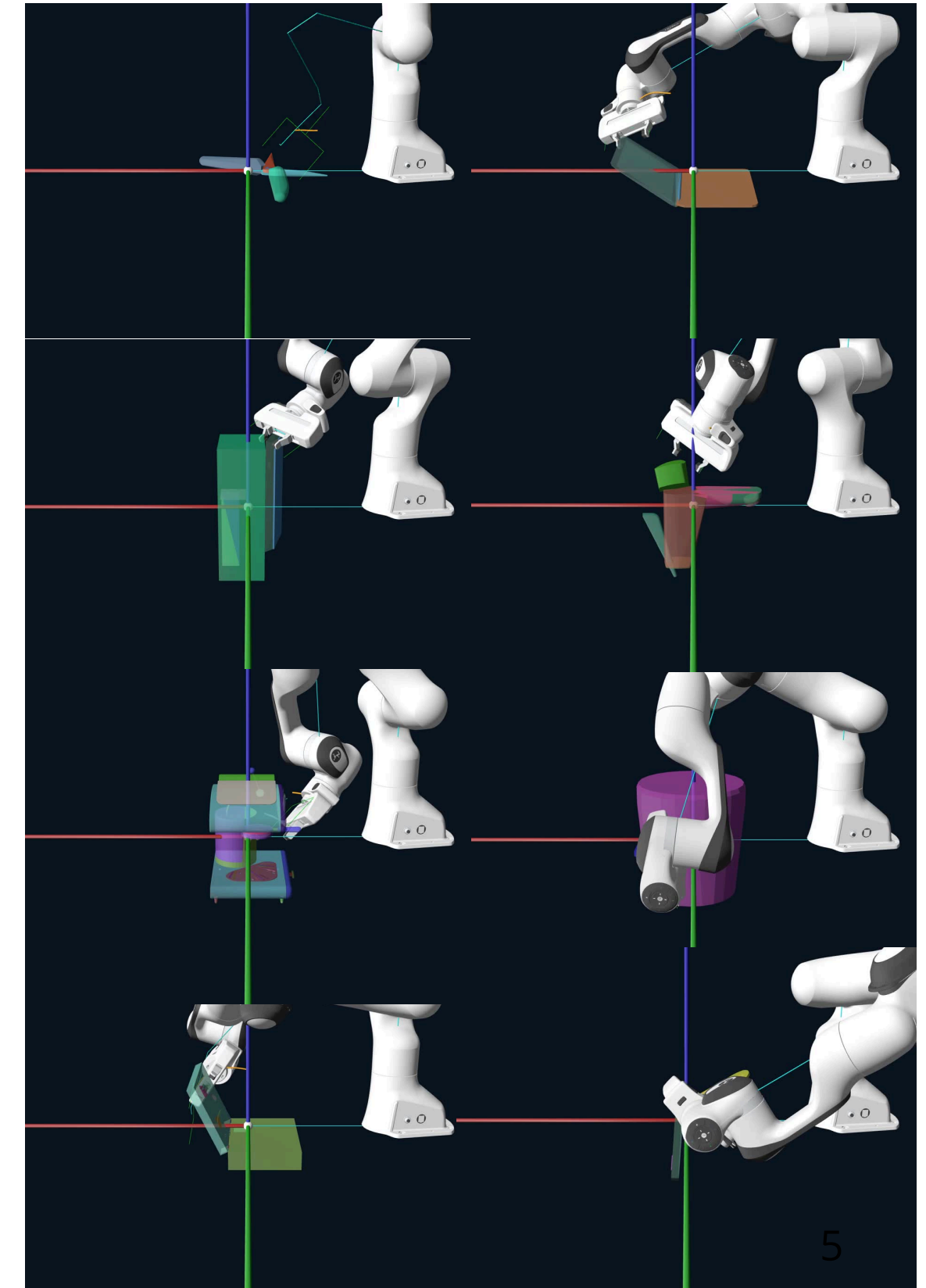
Automatic expert data generation through learning in simulation

QDTraj = Generation of expert trajectories in simulation + Real Robot deployment

$$\mathcal{T}_\Gamma = \left\{ \mathcal{F}_{ee/object}^{(k)} \right\}_{k=0}^N$$



Real-world Cartesian Path Following



QDTraj explores multiple ways to accomplish the same task

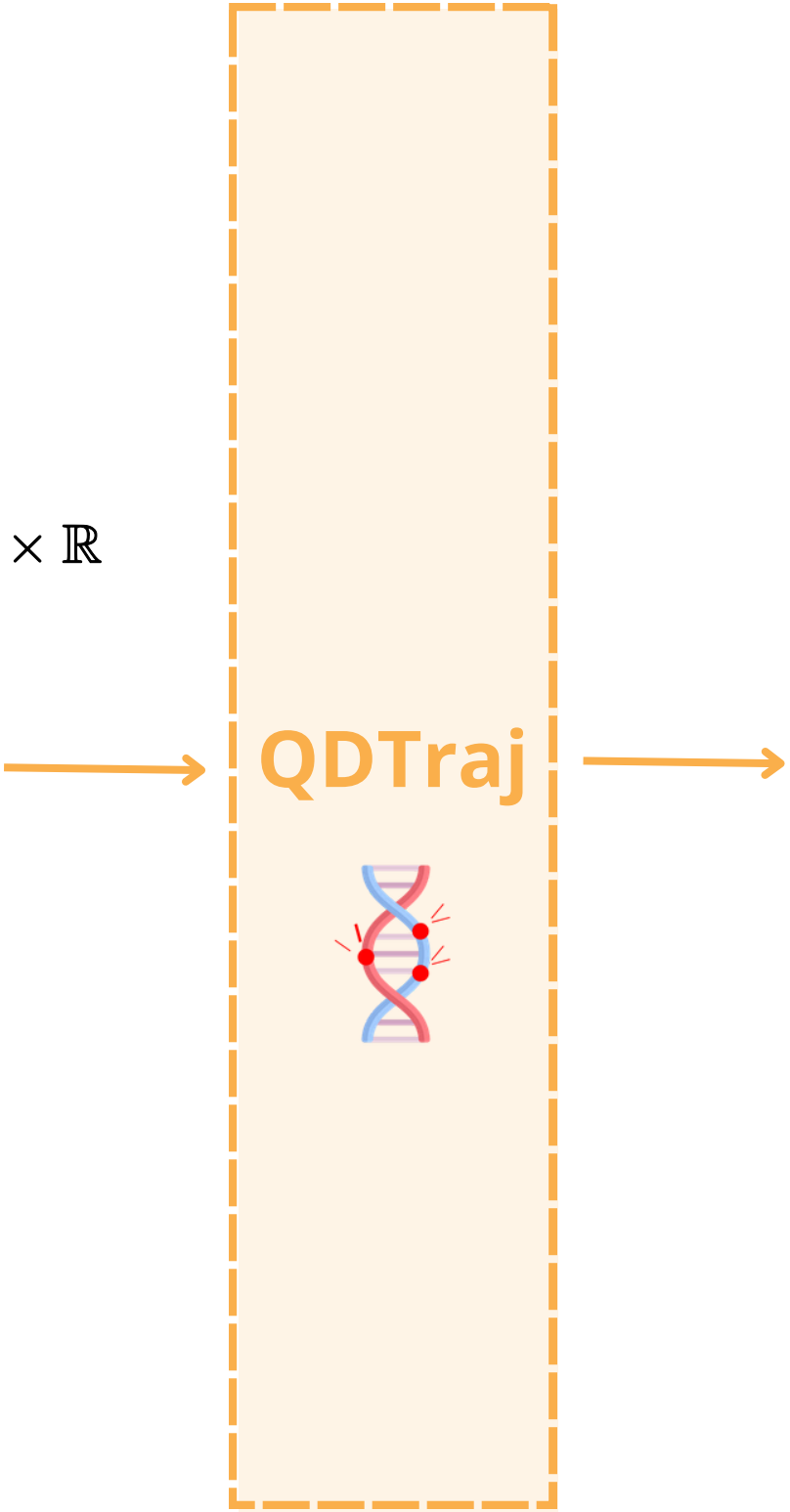
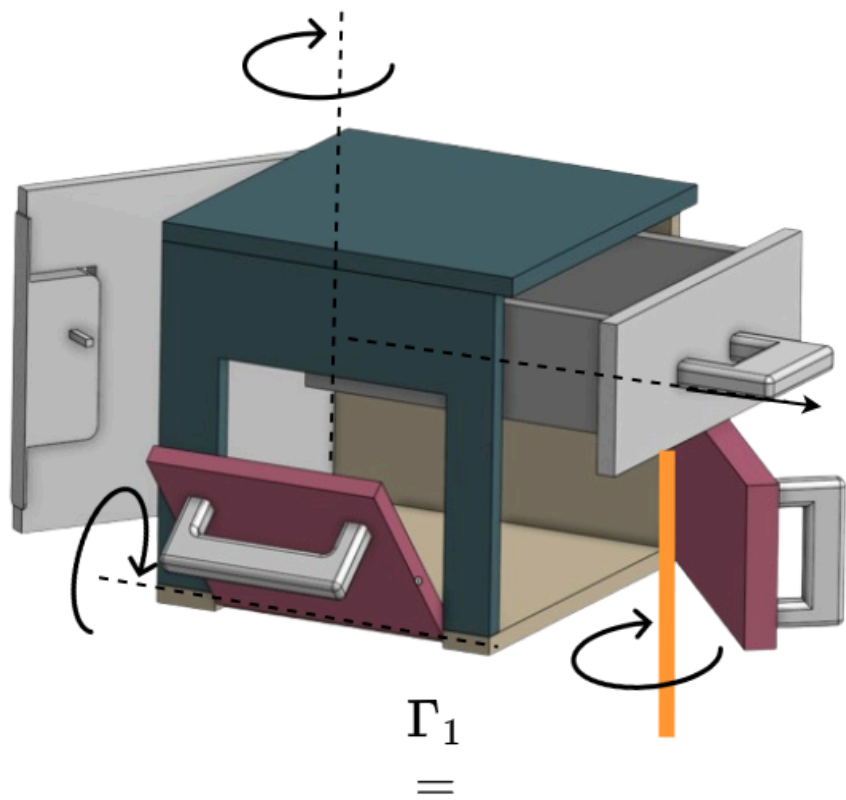
QDTraj Action Primitive representation

One Action Primitive

=

A robot manipulation task for activating
a single object articulation

$$\Gamma = (\text{object}, \text{articulation+index}, s_{\text{init}}, s_{\text{target}}) \in \mathcal{X} \times \mathcal{A} \times \mathbb{R} \times \mathbb{R}$$

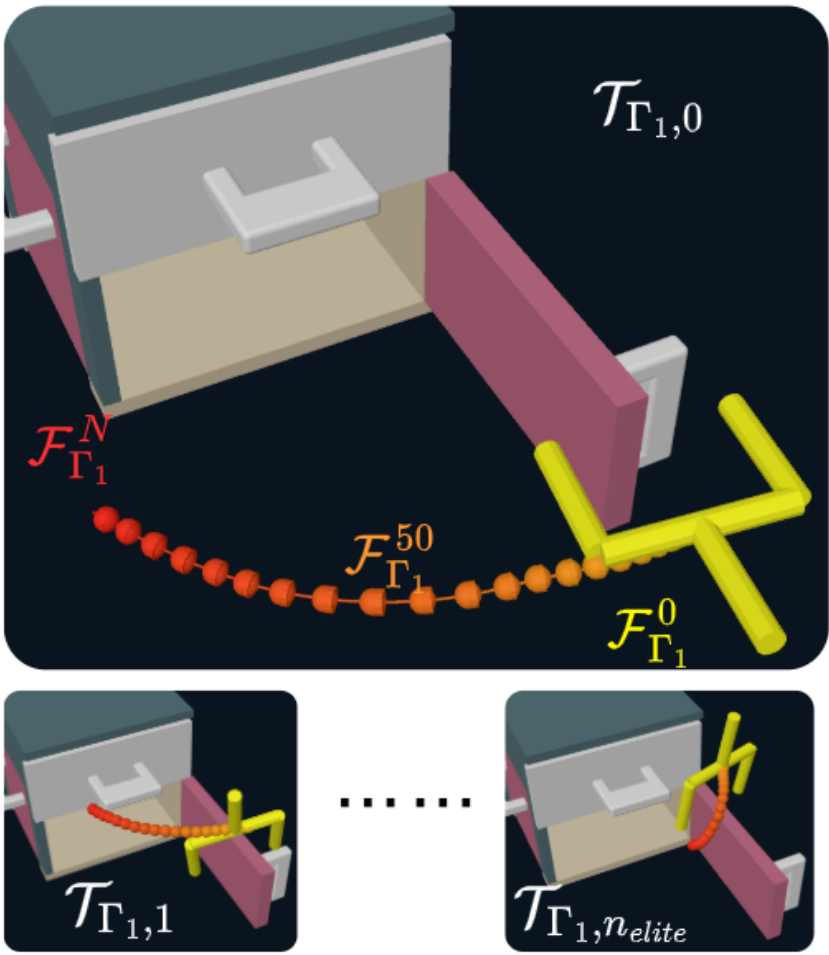


Set of Trajectory Primitives

=

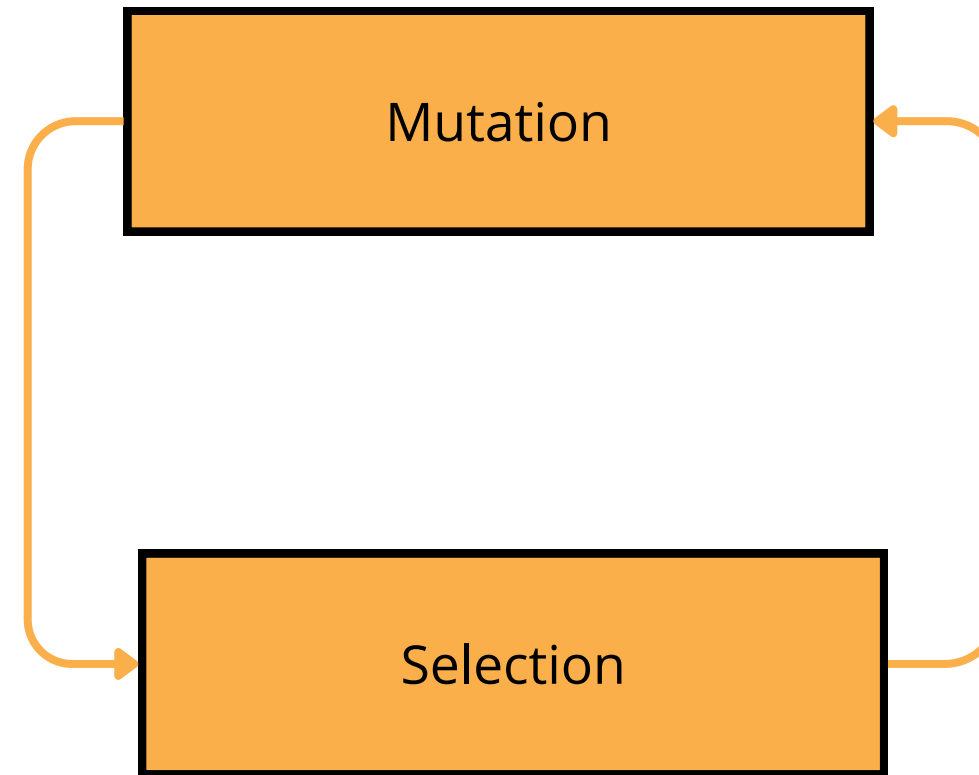
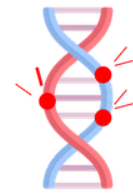
Sequence of trajectory key points
(6 DoF eef poses)

$$\mathcal{T}_{\Gamma} = \left\{ \mathcal{F}_{ee/object}^{(k)} \right\}_{k=0}^N$$



Learning Methodology relying Quality-Diversity

Quality-Diversity Algorithms



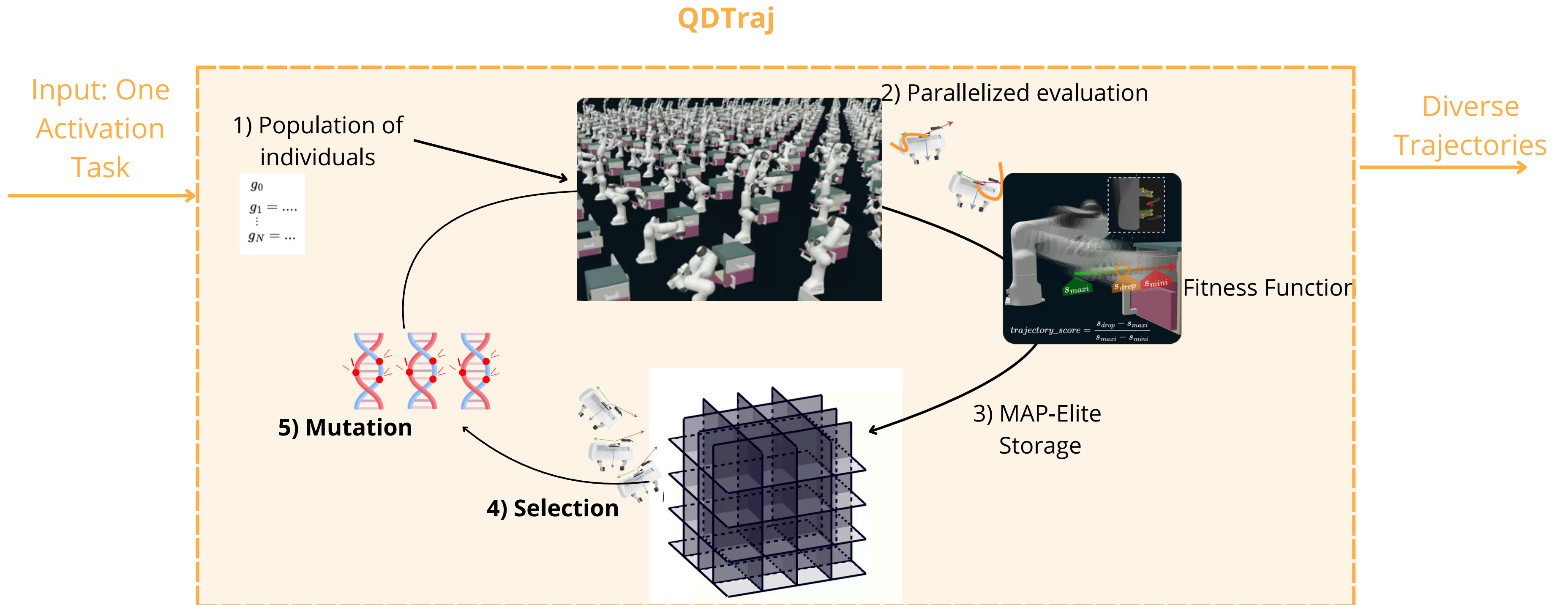
- Excellent exploration properties.
- Generate a repertoire of diverse and high-performing solutions. [1][2]
- Family of Evolutionary Algorithms.
- Selection based on MAP-Elites archive[3].

[1] J. Huber, F. Héli  n, **M. Kappel**, E. Chelly, M. Khoramshahi, F. Ben Amar, and S. Doncieux, "Speeding up 6-DoF Grasp Sampling with Quality-Diversity," IROS, 2024

[2] J. Huber, F. Héli  n, **M. Kappel**, I. de Loyola P    -Ubieta, S. T. Puente, P. Gil, F. Ben Amar, and S. Doncieux, "QDGset: A Large Scale Grasping Dataset Generated with Quality-Diversity," ICRA, 2025

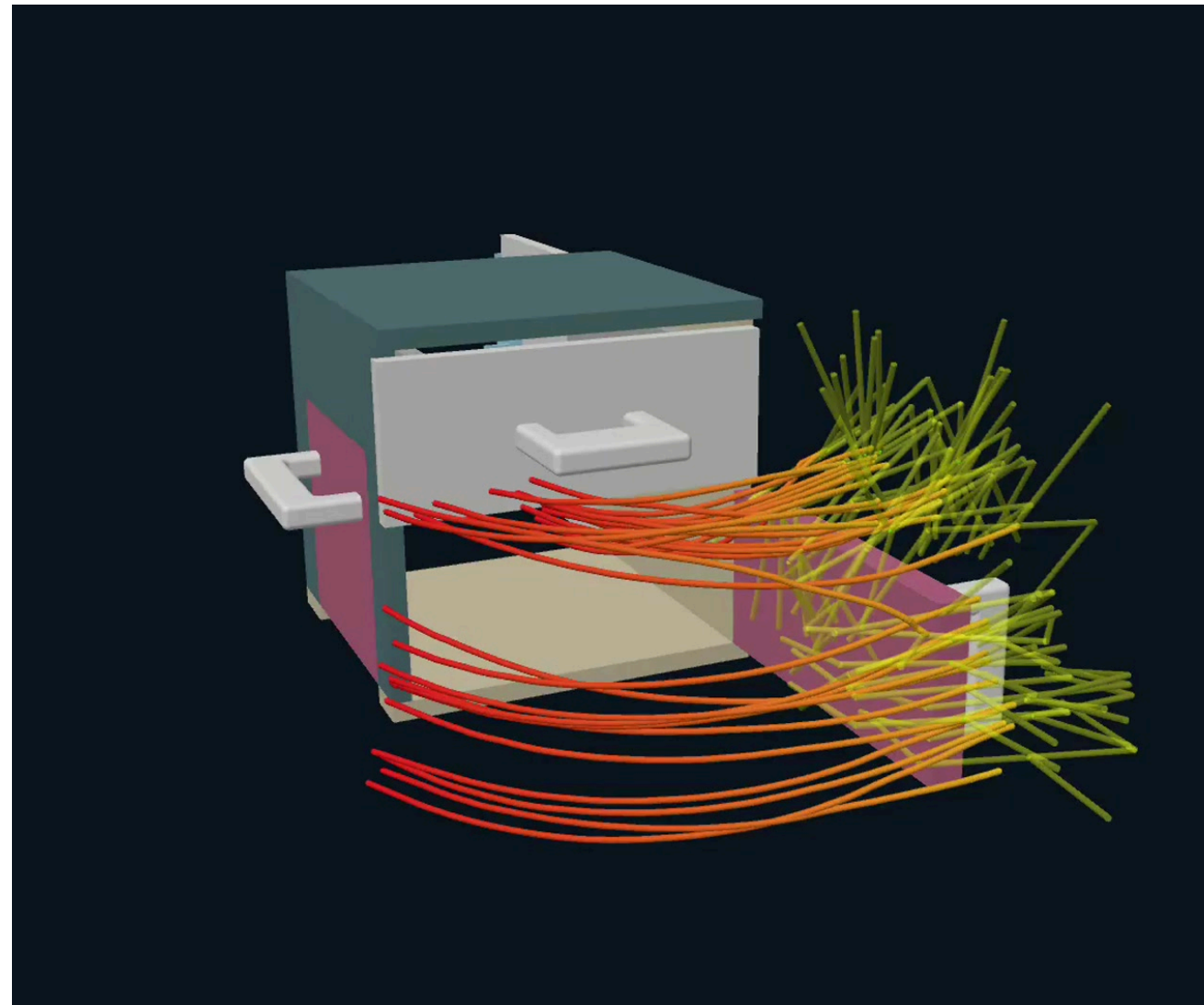
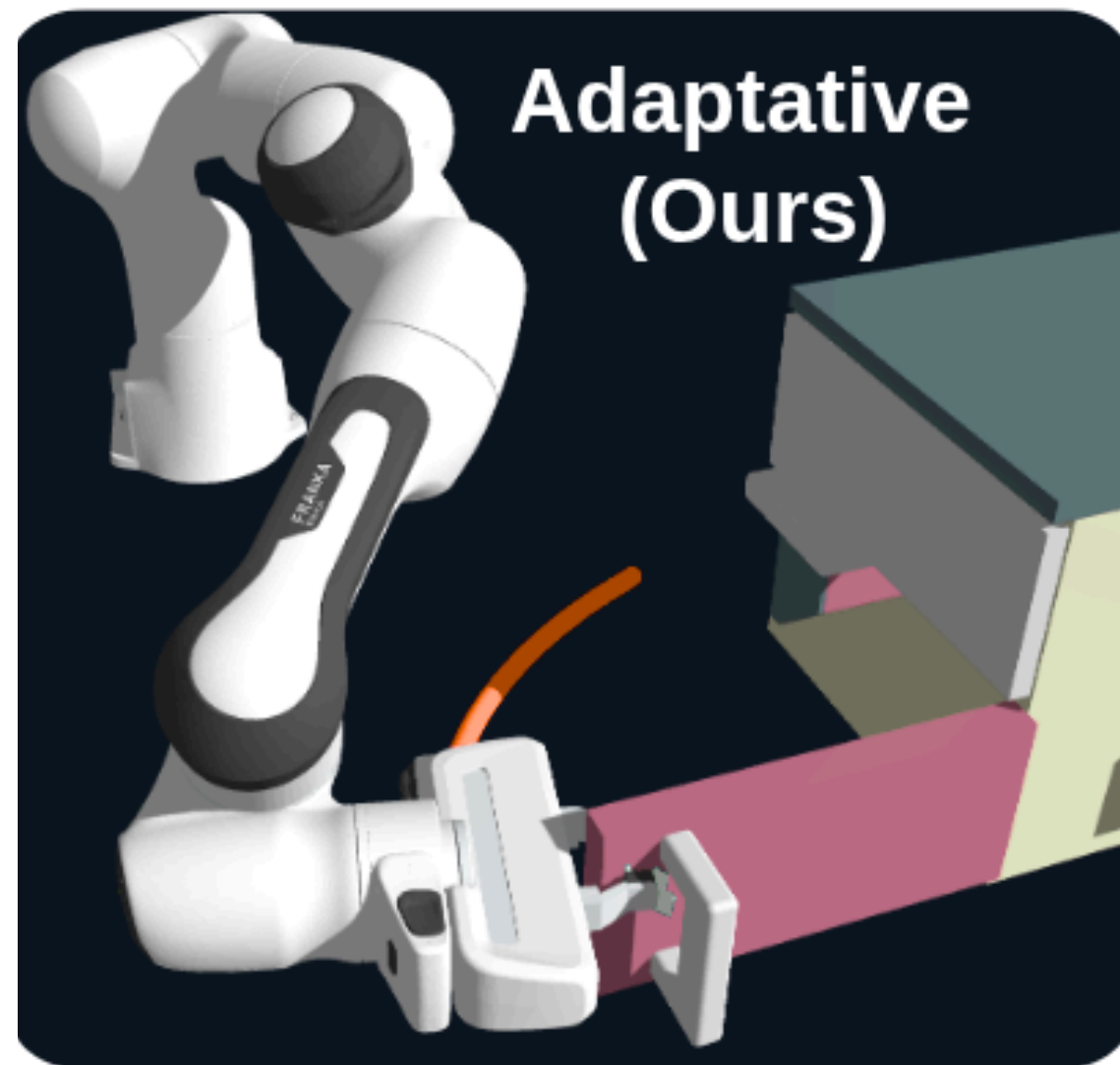
[3] J.-B. Mouret and J. Clune, "Illuminating search spaces by mapping elites," Apr. 20, 2015

QDTraj mutation-selection loop



Adaptive Interaction during evaluation process

- Adaptive: The robot is switched to compliant mode.
- **The object leads the robot.**

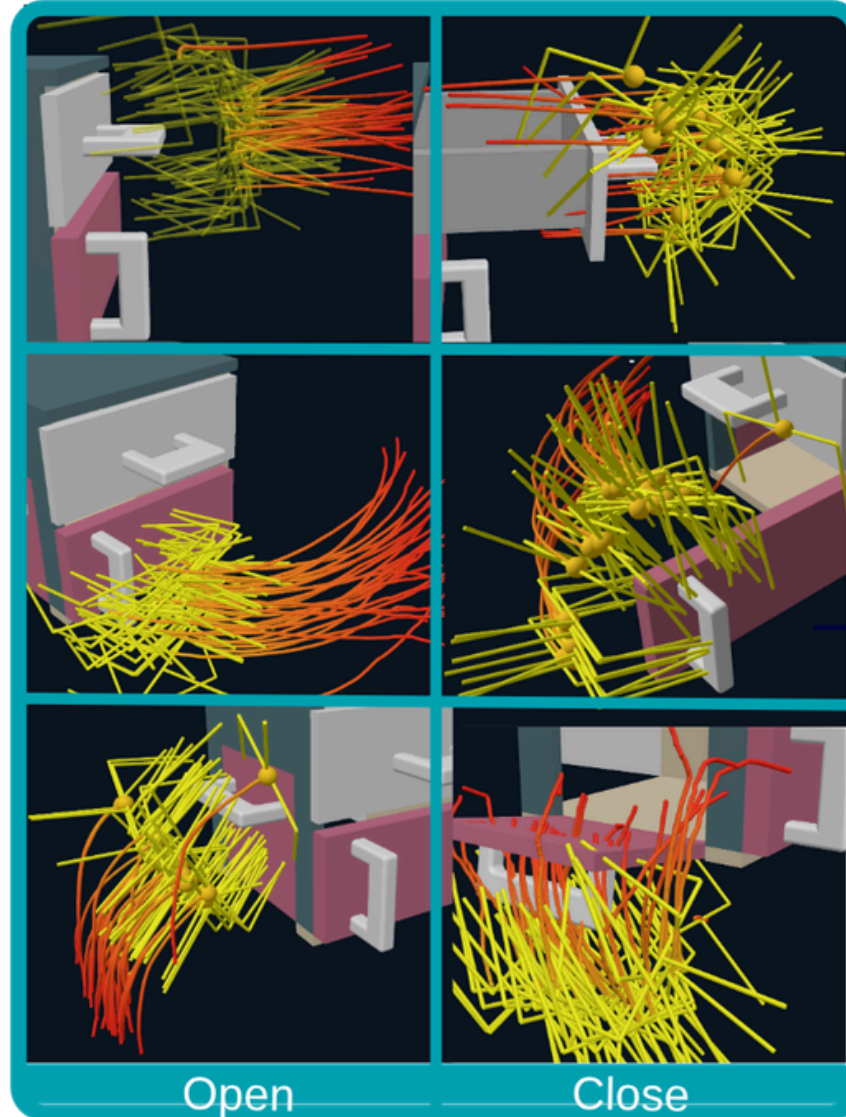


Robot : Low gains
Object : High gains

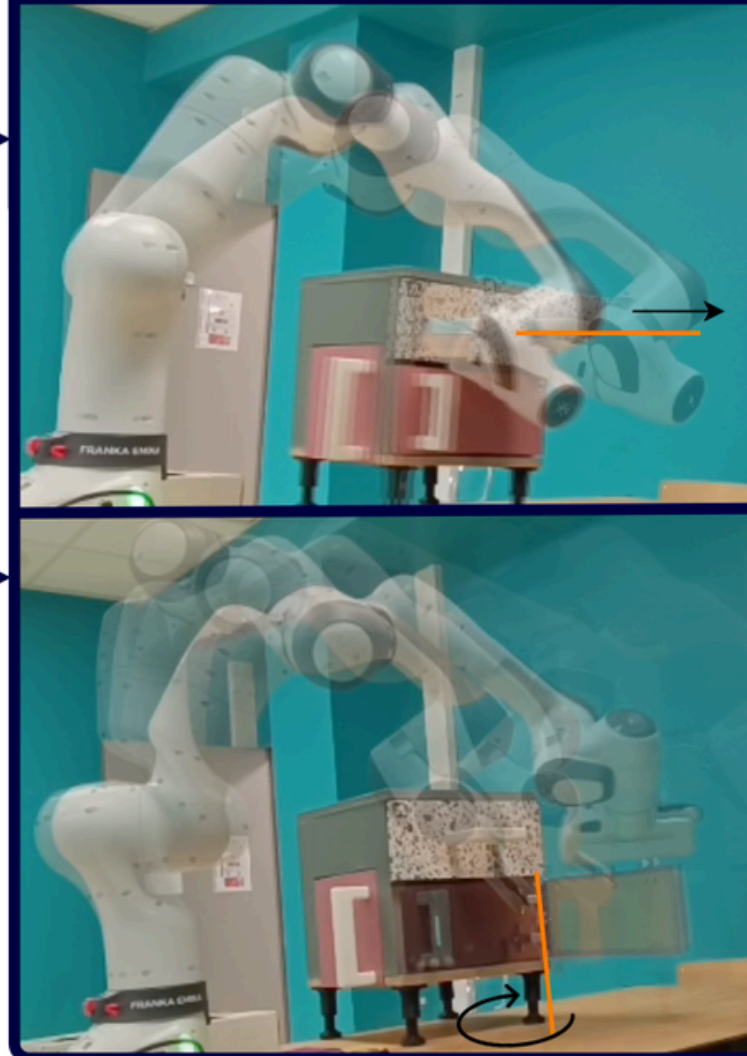


QDTraj Experiments

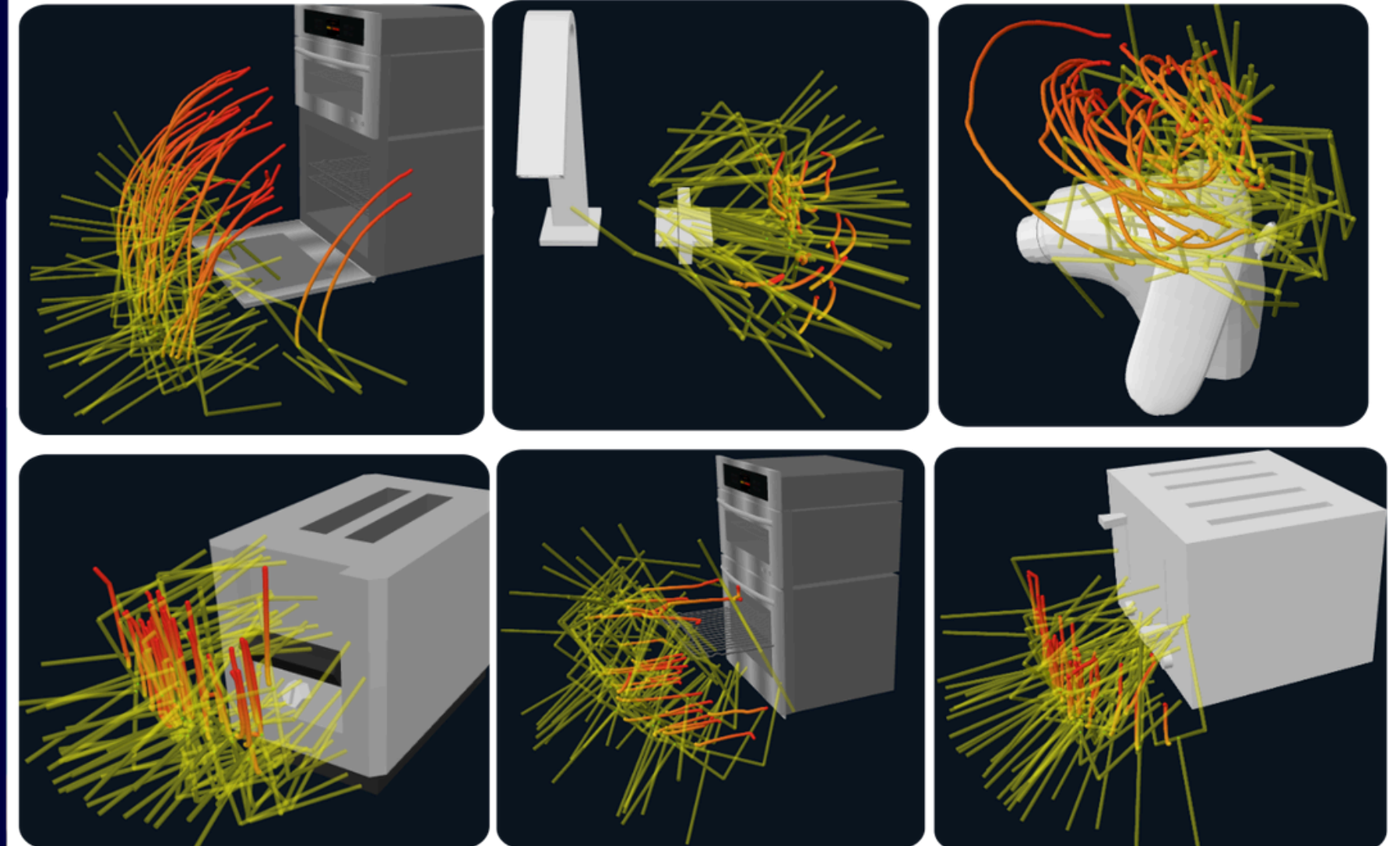
Trajectory Generation on Experimental Box



REAL ROBOT DEPLOYMENT



Trajectory Generation on PartNet Mobility Dataset



+ Comparison with other sampling strategies and interaction strategies

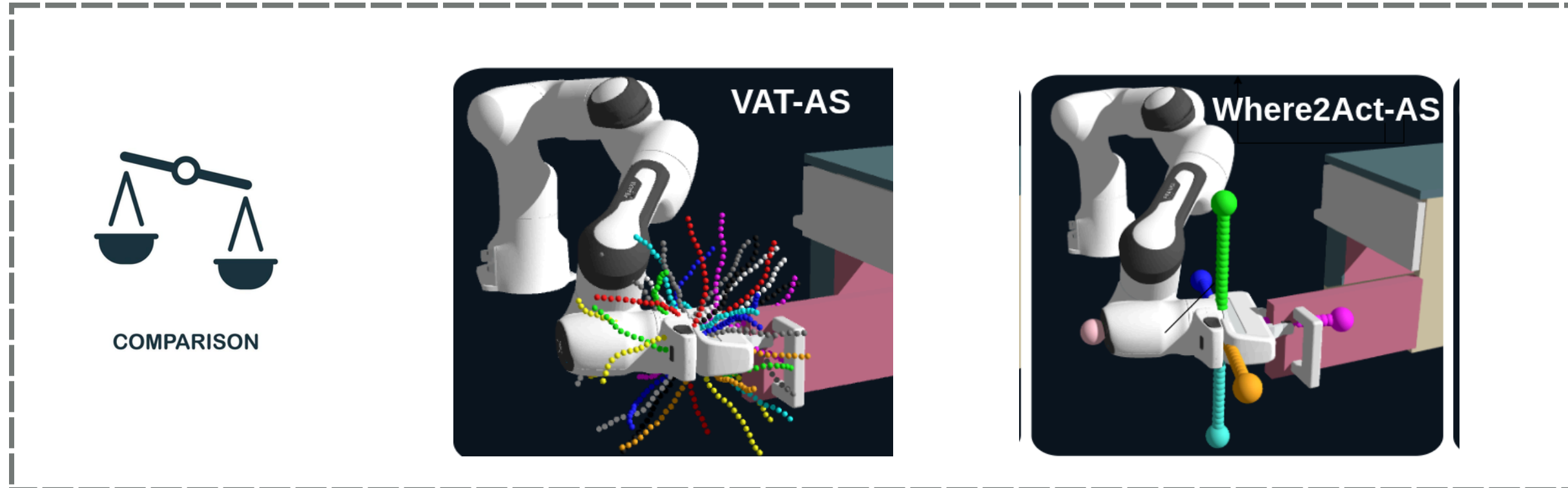
QDTraj Results

Learning strategy MAP-Elite-Explore:

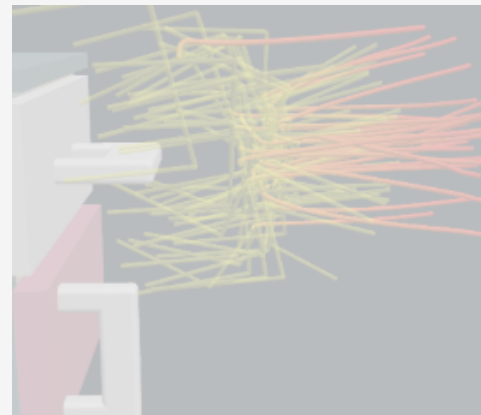
- **5 X** more successful trajectories than MAP-Elite-Success.
- **40 X** more successful trajectories than Random sampling.

Adaptive Action Space :

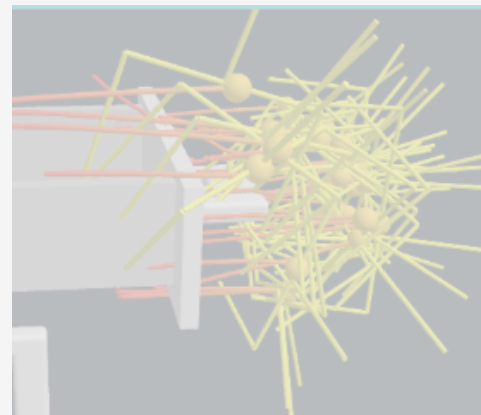
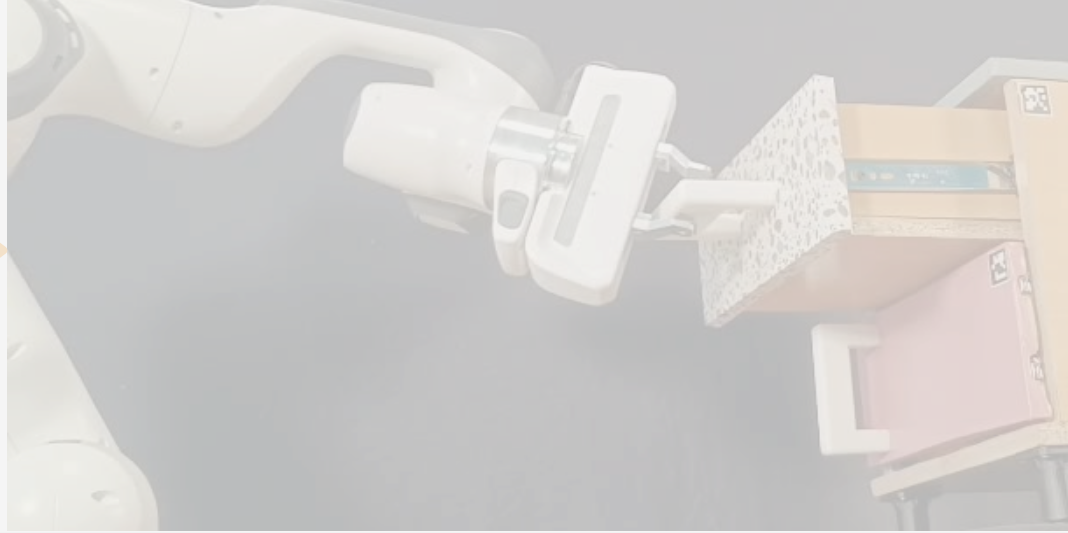
- **2.3 X** more successful diverse trajectories than using VAT-AS & Where2Act-AS.



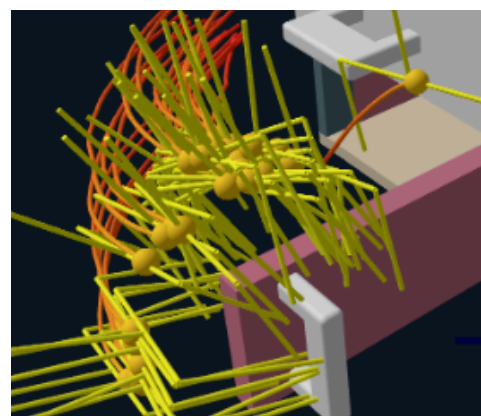
QDTraj real-world integration



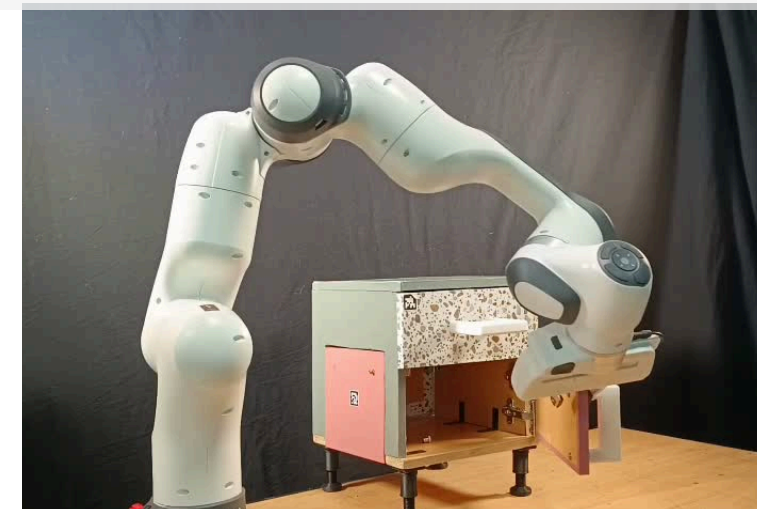
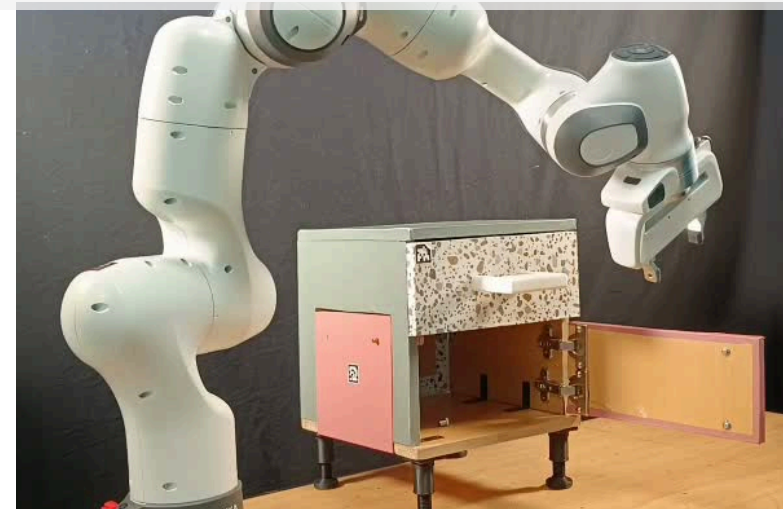
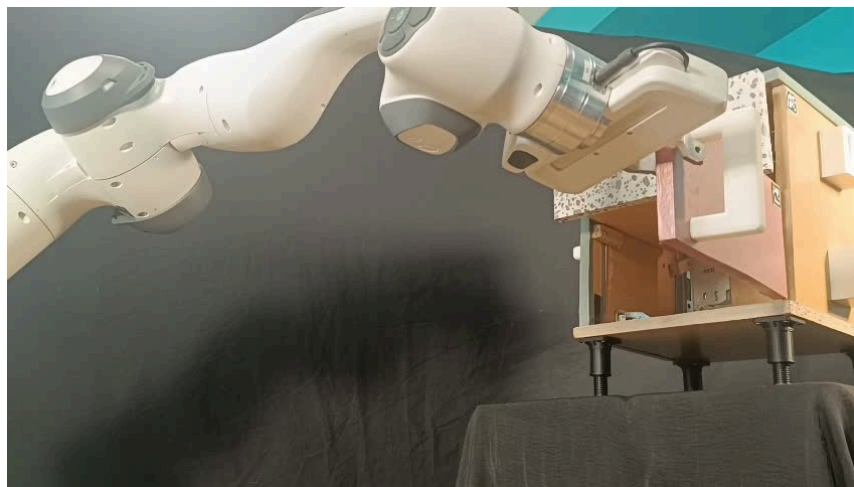
Open drawer



Close drawer



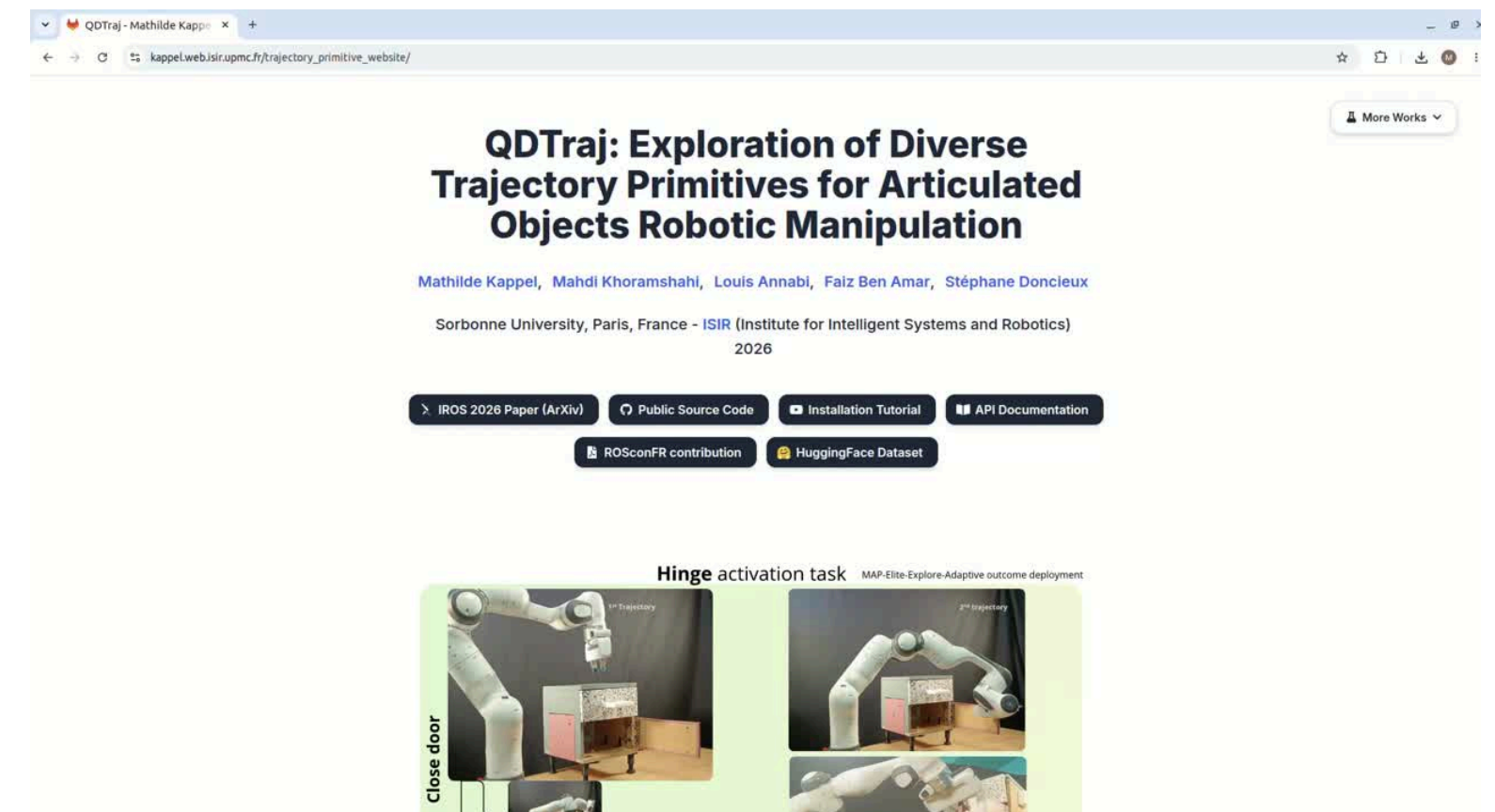
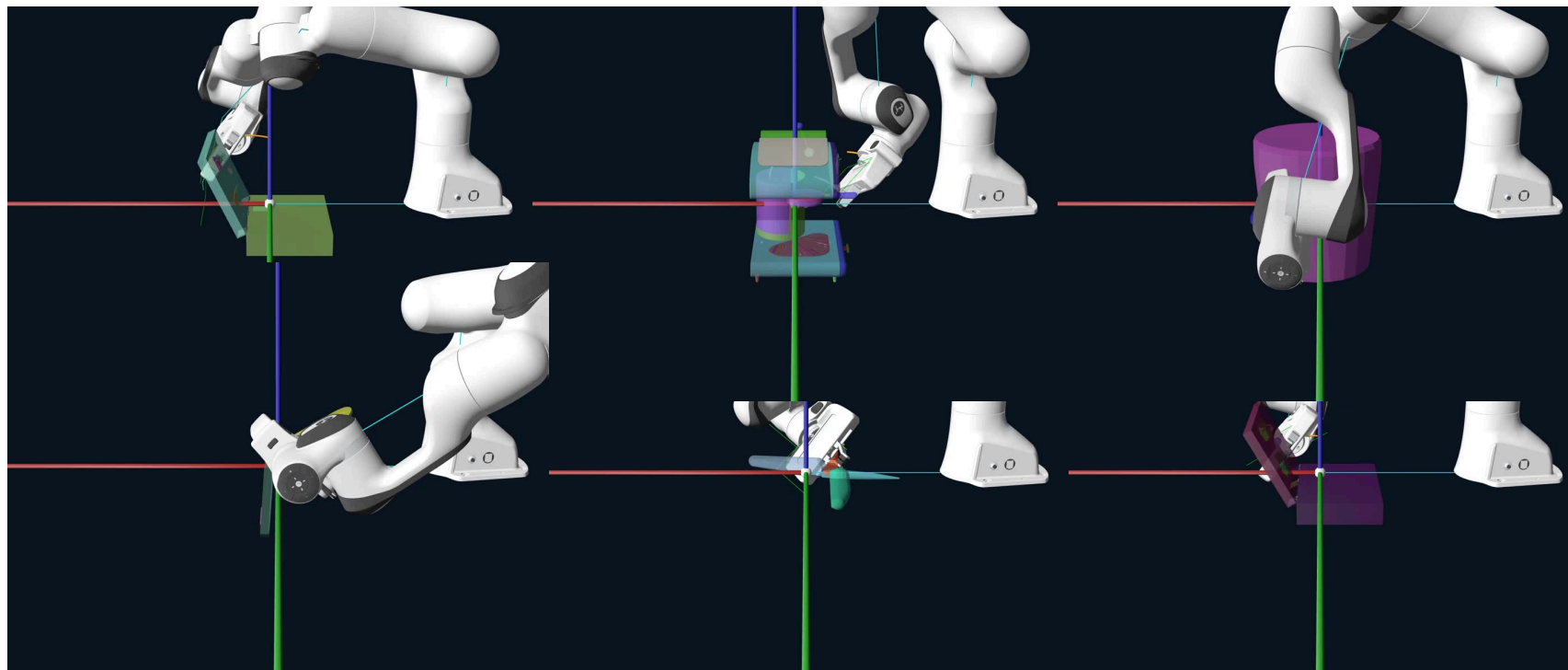
Close door



Results and Conclusion

Take-home message about QDTraj:

- QDTraj is a method for generating **diverse low-level expert trajectory primitives** to manipulate articulated objects.
- It leverages parallelized **Quality-Diversity algorithms**.
- It uses an adaptive interaction strategy relying on **compliant control**.



Project webpage: https://kappel.web.isir.upmc.fr/trajectory_primitive_website/

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Thank you



Poster Pod G3 East Face



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